

Mathematics A

Unit 2 • Chapter OL-T29 Classified

Cross-session • chapter-organised candidate

17 classified items • 71 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Unit 2 • Chapter OL-T29 • 17 items • 71 marks

Chapter	Items	Marks	Starts
01. OL-T29 Sequences	17	71	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Sequences

Unit 2 • 71 marks • 17 item(s)

June 2025 | 4MA1-1H Q24 | 5 marks

Recover first term and common difference

June 2024 | 4MA1-1H Q1 | 3 marks

Arithmetic nth term from listed terms

June 2024 | 4MA1-1H Q24 | 6 marks

Nonlinear or integer-constrained series recovery

November 2023 | 4MA1-1H Q19 | 6 marks

Recover number of arithmetic terms

June 2023 | 4MA1-1H Q20 | 6 marks

Recover number of arithmetic terms

November 2024 | 4MA1-1H Q19 | 5 marks

Recover first term and common difference

June 2022 | 4MA1-1H Q1 | 2 marks

Arithmetic nth term from listed terms

June 2022 | 4MA1-1H Q1 | 1 marks

Arithmetic term at a symbolic position

January 2022 | 4MA1-2H Q26 | 5 marks

Nonlinear or integer-constrained series recovery

January 2020 | 4MA1-2H Q25 | 3 marks

Recover first term and common difference

January 2021 | 4MA1-1H Q24 | 4 marks

Symbolic relation between partial sums

June 2021 | 4MA1-1H Q2 | 2 marks

Arithmetic nth term from listed terms

June 2021 | 4MA1-1H Q23 | 5 marks

Recover number of arithmetic terms

November 2021 | 4MA1-2H Q20 | 5 marks

Recover first term and common difference

May-Nov 2020 | 1H Q24 | 4 marks

Arithmetic nth term from listed terms

... 2 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

24 An arithmetic series has 30 terms.

The first term is a

The common difference is d

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of a and the value of d

Show clear algebraic working.

$a = \dots\dots\dots$

$d = \dots\dots\dots$

(Total for Question 24 is 5 marks)

WORKING SPACE

Question 01

4MA1-1H Q1

MODULAR UNIT 2

3 marks

- 1 Here are the first four terms of an arithmetic sequence.

1 4 7 10

- (a) Find an expression, in terms of n , for the n th term of this sequence.

.....
(2)

The n th term of a different arithmetic sequence is $5n + 17$

- (b) Find the 12th term of this sequence.

.....
(1)

WORKING SPACE

Question 11

4MA1-1H Q24

MODULAR UNIT 2

6 marks

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°

The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

○

WORKING SPACE

Question 15

4MA1-1H Q19 demand

MODULAR UNIT 2

6 marks

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x

The sum of the terms of the sequence is 7260

Find the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

WORKING SPACE

Question 16

4MA1-1H Q20

MODULAR UNIT 2

6 marks

20 The sum of the first 80 terms of an arithmetic series, S , is 470

The 75th term of S is 14.5

The sum of the first X terms of S is 171

Work out the value of X

Show your working clearly.

$X =$

WORKING SPACE

Question 12

4MA1-1H Q19 M01

MODULAR UNIT 2

5 marks

19 An arithmetic series has first term a and common difference d

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.
Show clear algebraic working.

.....

WORKING SPACE

Question 01

4MA1-1H Q1 Q1(a)

MODULAR UNIT 2

2 marks

- 1 Here are the first five terms of an arithmetic sequence.

 $1 \quad 5 \quad 9 \quad 13 \quad 17$

- (a) Find an expression, in terms of n , for the n th term of this sequence.

.....
(2)

WORKING SPACE

Question 02

4MA1-1H Q1 Q1(b)

MODULAR UNIT 2

1 marks

The n th term of another arithmetic sequence is $3n + 5$

(b) Find an expression, in terms of m , for the $(2m)$ th term of this sequence.

.....
(1)

WORKING SPACE

Question 38

4MA1-2H Q26

MODULAR UNIT 2

5 marks

- 26 An arithmetic series has first term a and common difference d , where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$

$$S_{2m} = 320$$

Find the value of d and the value of m

Show clear algebraic working.

$$d = \dots\dots\dots$$

$$m = \dots\dots\dots$$

WORKING SPACE

25 Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$ k

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of k .

$k = \dots\dots\dots$

YOUR WORKING

24 An arithmetic series has first term a and common difference d .

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d .

Show your working clearly.

$a = \dots\dots\dots$

YOUR WORKING

2 Here are the first 4 terms of an arithmetic sequence.

85 79 73 67

Find an expression, in terms of n , for the n th term of the sequence.

YOUR WORKING

23 The sum of the first N terms of an arithmetic series, S , is 292

The 2nd term of S is 8.5

The 5th term of S is 13

Find the value of N .

Show clear algebraic working.

$N = \dots\dots\dots$

YOUR WORKING

20 Here are the first four terms of an arithmetic series.

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

Given that the 15th term of the series is $(90 + 2k)$,

calculate the sum of the first 30 terms of the series.

YOUR WORKING

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

(Total for Question 24 is 4 marks)

YOUR WORKING

- 22 The first term of an arithmetic series is $(2t + 1)$ where $t > 0$
The n th term of this arithmetic series is $(14t - 5)$

The common difference of the series is 3

The sum of the first n terms of the series can be written as $p(qt - 1)^r$ where p , q and r are integers.

Find the value of p , the value of q and the value of r
Show clear algebraic working.



YOUR WORKING

$p =$ $q =$ $r =$

(Total for Question 22 is 4 marks)

YOUR WORKING

- 25** The first term of an arithmetic series is 10
The 20th term of the series is 86
- The sum of the first N terms of the series is 5194
- Work out the value of N
Show your working clearly.

$N = \dots\dots\dots$

(Total for Question 25 is 5 marks)

YOUR WORKING
